Mostafa Mahmoud

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About Me

A 3rd-year Mechatronics Engineering student at Yildiz Teknik University (GPA: 3.4) with a strong focus on embedded systems, automation, and real-time control. Skilled in Python, C++, Embedded C, SolidWorks, and micro-controller-based development (Arduino, PIC, STM32), with practical experience in Proteus simulation and 3D design. I enjoy transforming concepts into functional prototypes and have hands-on experience in robotics, control systems, and AI-driven applications. I am eager to contribute to cutting-edge projects through internships or part-time roles while deepening my expertise in embedded technologies and advanced control systems.

Education

Yildiz Technical University, Istanbul, Türkiye 2023 – 2027
B.Sc. in Mechatronics Engineering | Candidate at M2 lab | GPA: 3.35 *3rd Year Undergraduate*

Projects

Teknofest Air Defense Competition

- Developed the whole Embedded control system for the system using STM32 MCU.
- Implemented a closed loop cascaded PID Controller to precisely aim at targets.
- Designed UART communication between the MCU and a PC. and I2C to read Encoder inputs.
- Developed a FreeRTOS software for precise time management.

Teknofest FPV Drone Tracking competition (currently ongoing)

- Flight Controller Configuration and system integration
- Developed MAVlink communication between Raspberry Pi companion computer, ground station, and Pixhawk 6x to transmit telemetry, drone states monitoring, and autonomous commands.
- Implemented closed-loop gimbal control to track FPV drone precisely.
- Integrated optical flow and LiDAR sensors for drone position estimation without GPS.

Teknofest AI in Aviation Competition

- Developed a YOLO-based computer vision pipeline using Python and OpenCV for real-time object detection from video streams.
- Implemented a monocular camera position estimation algorithm for target localization.

3-floor Elevator Control System with PIC Micro-controller

- Programmed a multi-floor IR remote controlled elevator system using a 360° servo motor with LCD display.

Arduino Learning Projects

- 4.1. Line-Following Robot: Built a robot with Arduino and sensors for real-time line tracking.
- 4.2. Smart Wind Turbine: Designed a lightweight turbine that auto-adjusts direction to wind using potential sensors.

Skills

Technical Skills Embedded Systems, Real Time Systems, Microcontrollers programming,STM32, PIC, Robotics, Computer Vision, Object detection
Software and Tools: STM32CubeIDE, STM32CubeMX, Arduino, Proteus, MATLAB, SolidWorks, Embedded C, C++, Python
Soft Skills: Leadership, Communication, Problem solving, adaptability and quick learning, Teamwork

Certifications

Introduction to Microprocessors – ARM

C programming Bootcamp - The complete C language Course – Udemy

Embedded C – Coursera

Embedded Systems Bare-Metal Programming (STM32) – Udemy

Mastering STM32CubeMX 5 and CubeIDE - Embedded Systems – Udemy

FreeRTOS from Ground Up TM on ARM processors – Udemy

Extracurricular Activities

Private Tutor
Taught Math, Differential Equations, Physics, System dynamics, Signals, and Micro-controllers.

Volunteered as the Head of the Academic Committee
In a club in the university. Guiding students through their academic journey and preparing study plans for them.

Motor Control Workshop
A practical workshop on how to control servo and stepper motors using STM32.